FanStudio - World Cup 2026

Interactive Generative 3D Fan Experience

1. Project Overview

FanStudio is a high-performance web application designed for the FIFA World Cup 2026. It allows football fans to select their favorite national team, take or upload a smartphone selfie, and instantly generate hyper-realistic, branded 2D photos. From there, fans can transform their favorite generation into an interactive, volumetric 3D spatial asset (Gaussian Splat) that they can view, tilt, and interact with in real time.

The Core Value Proposition

- **Zero App-Store Friction:** A lightweight, mobile-responsive web experience.
- **High-Fidelity Identity Keeping:** Blends the user’s true facial structure seamlessly into official team kits.
- **Cutting-Edge Spatial Tech:** Bridges the gap between 2D AI generation and 3D web-graphics pipelines in seconds.

2. Core Features & Technical Architecture

A. High-Fidelity Outfit Integration (SAM 3 + IP-Adapter)

- **Segment Anything (SAM 3):** Automatically isolates jerseys, shorts, and specific team apparel from high-resolution reference lookbooks, stripping out backgrounds or promotional models cleanly.
- **IP-Adapter-Flux regional conditioning:** Feeds the isolated fabric texture, emblems, and colorways precisely into the target composition while completely masking off the head/neck region to preserve the user's facial identity.

B. Serverless AI Orchestration (Modal + ComfyUI API)

- **FLUX.2 Klein (9B Distilled):** Serves as the primary inference engine. The distilled framework cuts generation latency dramatically while preserving complex prompt adherence (stadium floodlights, cinematic fabric textures, crowds, bokeh).
- **Apple SHARP (Sharp Monocular View Synthesis):** Powers the 3D pipeline. Rather than calculating dense, slow polygon meshes, SHARP takes a single 2D image and maps its depth and view-dependent radiance into a 3D Gaussian Splat (`.ply` format) via a rapid feedforward pass on the backend.
- **Decoupled Multi-Container Pipeline on Modal:** Inference is broken into two standalone stages to avoid billing for unnecessary 3D compute and to keep user-perceived waiting times low.

C. Client-Side Volumetric Rendering

- **WebGL/WebGPU Splatting:** The frontend fetches the raw `.ply` file generated by Modal and renders the 3D Gaussian Splat directly on the phone's GPU using WebGL. No expensive server-side video rendering or heavy streaming is required.

3. The User Experience (UX) & App Flow

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[ Step 1: Landing ] → [ Step 2: Camera ] → [ Step 3: 2D Generation ] → [ Step 4: 3D Studio ]
  Select Team            Capture Selfie         View 4 FLUX Variances          Tilt & Interact
```

Step 1: Team Selection & Landing

- **The Screen:** A clean, minimalist dark-mode interface displaying an indexed grid of World Cup 2026 qualified teams.
- **The Action:** The user taps on their favorite country (e.g., Argentina, Egypt, Brazil, Mexico).
- **The Uniform:** The app defaults strictly to the 🏠 **Home Kit** configuration for all teams to maximize instant visual recognition and keep asset delivery streamlined.

Step 2: Identity Capture

- **The Action:** The user grants camera permissions to take a direct selfie with their phone's front camera, or selects a photo from their camera roll.
- **UX Guardrails:** A simple on-screen silhouette overlay guides the user to center their face, ensuring optimal alignment for the face-swapping nodes.

Step 3: The 2D Generation Grid (The FLUX Stage)

- **The Transition:** While Modal scales up the container, the frontend displays contextual tailor-themed progress states (e.g., *"Tailoring your jersey..."*, *"Positioning stadium lights..."*).
- **The Result:** The screen populates with a clean 2×2 grid showing **4 unique cinematic photos** of the user fully kitted out in their team's uniform inside a packed tournament stadium.
- **Action Call:** The user can download any of the flat 2D images directly or select their single favorite shot to hit the prominent **"View in 3D"** button.

Step 4: The 3D Spatial Canvas (The SHARP Stage)

- **The Transition:** The app triggers the second, isolated Modal container running the Apple SHARP workflow, flashing a progress indicator (*"Generating volumetric space..."*).
- **The Interactive Canvas:** The flat photo disappears, replaced by an interactive 3D scene. The user can swipe left/right/up/down to orbit around themselves or tilt their physical smartphone to use device gyroscope data, giving the image an immersive, holographic depth effect.

4. Frontend Engineering Specification

Recommended UI & Graphics Stack

- **Framework:** Next.js (React App Router) for structuring routing and components, deployed over a global edge network to ensure immediate page loads and secure API keys.
- **Styling:** Tailwind CSS for a highly responsive, modern, dark-mode design system.

- **3D Viewport Layer: React Three Fiber (R3F)** combined with the `@react-three/drei` helper library.
    - *Implementation:* Utilize the built-in `<Splat />` component inside a standard Three.js canvas. This allows the `.ply` file to load natively as a point-cloud volumetric object with smooth camera controls and immediate mobile touch/gyro responsiveness.

5. Next Steps & Developer Implementation Milestones

1. **Phase 1: Modal Infrastructure Setup**
    - Deploy the FLUX.2 Klein 9B Distilled image script to Modal.
    - Deploy the Apple SHARP 3DGS custom node image script to Modal as a separate function.
    - Benchmark response times and adjust container auto-scaling boundaries.
2. **Phase 2: Frontend Construction**
    - Build out the responsive team selection, camera handling wrapper, and the Next.js API route handlers to hide backend tokens.
3. **Phase 3: 3D Canvas Integration**
    - Set up the React Three Fiber viewport using `@react-three/drei` to render the incoming point cloud.
4. **Phase 4: Asset Ingestion**
    - Run your completed SAM 3 script over the lookbook references to isolate the home kits for your production image database.